JANUARY - JUNE 2024

GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

"To prepare human resources with value-based competency for technical advancements and growth of society."

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**



ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standards of the society.**
- **To prepare engineers with effective communication skills, professional ethics and leadership qualities.**



NEWSLETTER

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HOD'S MESSAGE



PROF. SANJAY D. JOSHI

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

PROF. SANJAY D. JOSHI
HOD EC Department
Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

ELECTRONICS AND COMMUNICATION ENGINEERING

Faculty Members



PROF. SANJAY D. JOSHI
DESIGNATION: ASSOCIATE PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM, IMAGE PROCESSING, MICROPROCESSOR – MICROCONTROLLER



DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP



PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS



DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS



PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION



PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY



PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN



PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS



PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR



L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L-203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.



ELECTRONICS AND COMMUNICATION ENGINEERING

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench



L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.



L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)



National Youth Festival

Event Date : 12th to 16th January 2024
Venue : Nasik, Maharashtra

National Youth Day in India, celebrated on January 12th, honors Swami Vivekananda's ideals to inspire the youth towards a better future. Initiated by the Indian Government in 1984, its objective is to instill national pride and educate youngsters about their rights. The day is marked by various activities, including the National Youth Festival since 1995, promoting cultural exchange and development. In 2024, the festival was held in Nasik, Maharashtra, showcasing the region's rich cultural heritage from January 12th to 16th.

The National Youth Festival 2024, held in Nasik, Maharashtra from January 12th to 16th, saw participation from over 18,000 youths representing different states and union territories, including NSS, NYKS, and volunteers. Two studenTS 1) Joshi Ashray (CSE 6th Semester) & 2) Poorva Ravat (CSE 6th Semester) from GTU were selected to represent the institute. Inaugurated by Prime Minister Narendra Modi, along with other dignitaries, the event focused on motivating youth towards Vikshit Bharat 2047 and government schemes. The day concluded with a lively concert by Bollywood singer Niti Mohan, leaving participants with cherished memories and positive energy.



Day - 1

Day 2 of the event kicked off with a positive and enthusiastic start, featuring a Suvichar (Youth Convention) Session. Shri Vilas Shinde shared insights on effective farming techniques, emphasizing the importance of agriculture for India's development. Prateek Khandelwal inspired attendees with his journey from a life-altering accident to founding Ramp My City, a startup dedicated to making public spaces accessible for the physically handicapped. The day concluded with a lively music concert by the Street Bus Singers team, leaving participants energized and inspired.



Day - 2

On day three of the festival, we began with high energy, starting with the "Lakshya Geet" and two motivational sessions led by industry experts. Participants were encouraged to strive for excellence and contribute to the nation's development and the goal of Viksit Bharat 2047. The day also featured expo stalls showcasing government schemes for startups and new inventions. Highlights included innovative technologies like the PVC pipe salt water welding machine and the High Altitude GPS Detection System. In the evening, a cultural show entertained everyone, leaving them with cherished memories and newfound knowledge.



Day - 3

Day four commenced with high spirits, starting with the uplifting "Lakshya Geet" and insightful sessions by esteemed figures empowering youth to contribute to national development. Cultural stalls displayed diverse products reflecting India's rich heritage, while food stalls offered a taste of the nation's culinary diversity.

The day concluded with participants departing enriched with memories and learnings.





Day - 4

The final day of the 27th National Youth Festival began with the customary "Lakshya Geet," evoking both energy and sadness as participants prepared to part ways. The valedictory event featured speeches by dignitaries, including the Sports & Youth Affairs Minister of Maharashtra, followed by prize distribution for state volunteer competitions. Emotions ran high as participants pledged to contribute to India's development by 2047, concluding the camp with lasting memories and lessons.





પાટણ સરકારી ઈજનેરી કોલેજના સ્વયંસેવકોની મહારાષ્ટ્રમ
આયોજિત રાષ્ટ્રીય યુવા મહોત્સવ ૨૦૨૪માં પસંદગી

Day - 5

Republic Day

Event Date : 26th January 2024

Venue : Administrative Block

On 26th January, GEC Patan's NSS unit celebrated the Indian Constitution coming into effect, which holds the record for being the world's longest written constitution. The event began with the Princi_al hoisting the .lag and the National Hnthem. H NSS volunteer delivered a s_eech on the Constitution' hi torical journey, followed by a patriotic song by male volunteers. The Student Coordinator highlighted the Sahaj Foundation _roject and the Sainik Welfare Fund Female NSS volunteers showcased their talent through a dance performance, and the event concl.ded with thanks .rom the NSS St.dent Coordinator.







PM's Virtual Meet

Event Date : 13th March 2024

Venue : EE & CSE Dept. Seminar hall

On March 13, 2024, GEC Patan's NSS unit hosted a special event to watch Prime Minister Narendra Modi participate in the "India's Techade: Chips for Vikshit Bharat" program via live YouTube session. The event began with welcoming the principal, professors, and students. At 10:30 a.m., the live session started with 3 lakh viewers from 470 districts and 60,000 educational institutions. PM Modi inaugurated three semiconductor projects worth 1.25 lakh crores via video conferencing, marking a significant milestone for India's future. He also launched the PM-SURAJ portal for social upliftment and employment and distributed Ayushman Health Cards and PPE kits to Safai Mitras. Modi emphasized the importance of inclusive development for achieving Vikshit Bharat@2047. The event ended with a sense of inspiration and concluded with the National Anthem.









Sainik Welfare Fund

Event date : 07th December 2023 to 20th March 2024
Venue : Patan City

The Sainik Welfare Fund initiative at Government Engineering College, Patan, in 2025, showcased the collective spirit of students and faculty towards honoring the sacrifices of ex-servicemen and the families of martyred soldiers. Guided by their professors and led by the NSS unit, volunteers actively collected donations within the college, receiving full support from Principal Dr. A.B. Dhruv and all departments. The activity not only strengthened students' values of service, teamwork, and patriotism but also extended beyond the campus, reaching government offices like the Collector Office, Mamlatdar Office, Nagarpalika, Police Stations, and several schools, where they spread awareness and gathered widespread support from officers, staff, and the general public. Soldiers play a crucial role in safeguarding our nation's peace and freedom, deserving our utmost respect and support. To aid their families, our university annually issues a fundraising guide. In 2024, NSS volunteers from GEC Patan continued their tradition by collecting donations from various sources, totaling **₹1,21,121**, the highest in eight years and in Gujarat State. This contribution was presented to the Vice-Chancellor of Gujarat Technological University on March 21, 2024, recognized as the largest among all affiliated NSS units for the year.





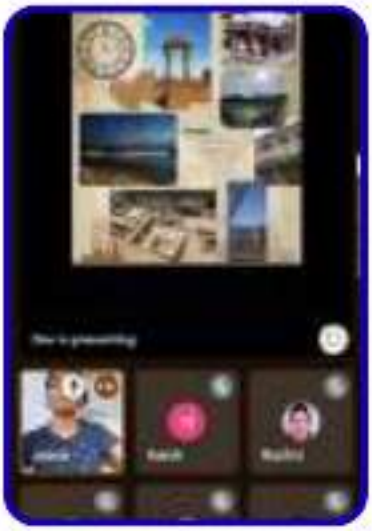
Gujarat Sthapana Divas

Event Date : 1st May 2024

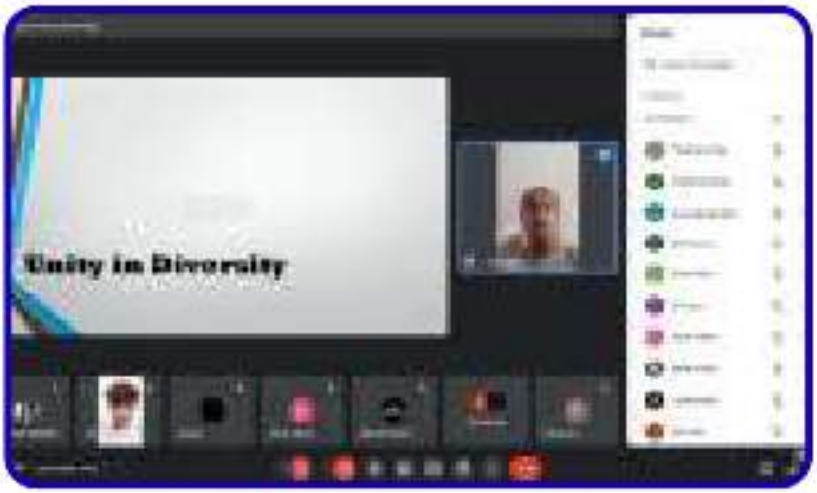
Venue : Google Meet (Virtual)

On the 64th Gujarat Sthapana Divas, NSS GEC PATAN organized an online quiz competition and an event titled "Gujarat's Past in Frames: Historical Image Description Contest." Participants presented and described historical places in Gujarat. Highlights included Manish Yadav's speech on Gujarat's importance, Harsh Yadav on the "Statue of Unity," Parmar Prerna on "Vadnagar village," Mal Jyoti on the "EME Temple," and Sharad Mishra on "Role of Gujarat in Vikshit Bharat @2047." The event concluded with encouragement for future participation.









Thalassemia Minor Screening Camp

Event date : 17th May 2024
Venue : Mechanical Workshop

Government Engineering College Patan, in collaboration with the Indian Red Cross Society, organized a Thalassemia Minor Screening camp, which focused on testing all students for this blood disorder. Thalassemia is a genetic condition that affects the body's ability to produce Hemoglobin. Early detection through such screening tests can help identify those who carry the Thalassemia gene. By getting tested, individuals can take preventive measures to protect their health.

The camp took place in the Mechanical Workshop at GEC Patan, starting at 10 a.m. A dedicated medical team was present to collect blood samples from the students. Beyond testing, the camp also had an educational aspect, where students were informed about Thalassemia, its various types, and the significant impact it can have on a person's life. The medical team stressed the importance of Thalassemia testing and raising awareness about the disorder.

To ensure a smooth process, time slots were assigned to different departments for the collection of blood samples, which helped in maintaining order and efficiency. The event had a strong turnout, with 219 students from various departments and semesters participating. The camp not only offered valuable knowledge about Thalassemia but also fostered a sense of community and unity among the participants. The event concluded with a group photo, symbolizing collective efforts in raising awareness for this crucial cause.





International Yoga Day

Event Date : 21st June 2024

Venue : GEC Patan Campus

NSS GEC Patan organized a yoga session at 7:00 a.m. in front of the Administrative Block for professors and students, led by yoga trainer Dewanshi Goswami. The session began with prayer, meditation, and warm-up exercises. Principal D.K. Patel, faculty member Sachin, HOD of Electrical Department G.A. Patel, and NSS Program Officer N.K. Dabhi emphasized the importance of yoga. Dewanshi Goswami led the participants through asanas designed to enhance well-being. The session concluded with G.A. Patel expressing gratitude to Goswami and presenting her with a Shrimad Bhagwat Gita as a token of appreciation.









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